

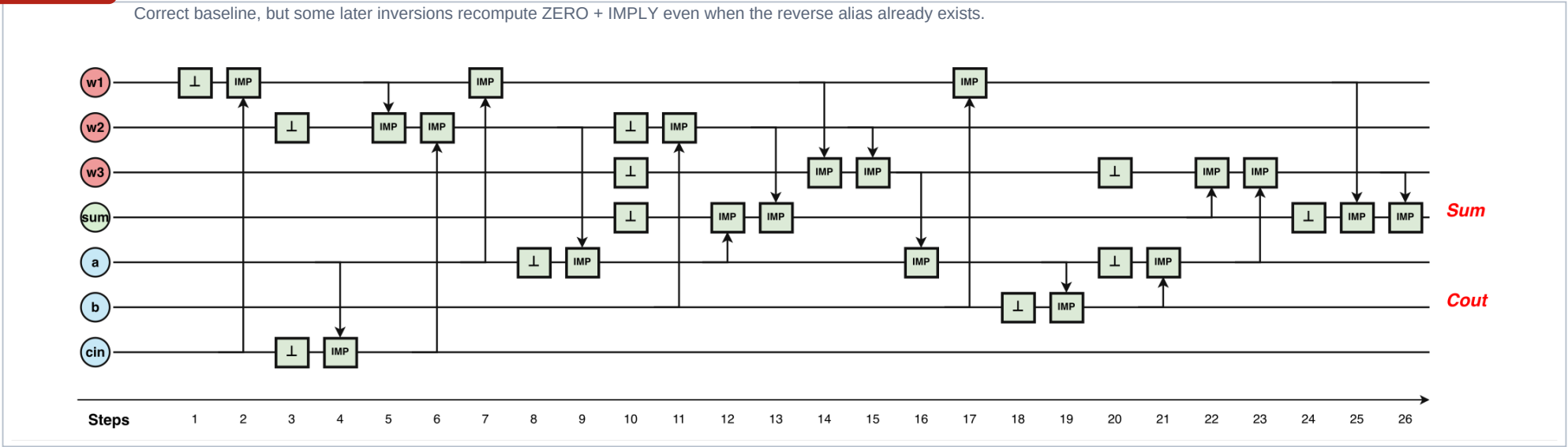
Full Adder Scheduling: 26 Steps -> 20 Steps

Same ABC mapping target; improvement comes from scheduling and alias bookkeeping after ZERO/IMPLY lowering.

BEFORE

26 steps: 7 FALSE pulses + 19 IMPLY pulses

Correct baseline, but some later inversions recompute ZERO + IMPLY even when the reverse alias already exists.



What changed

1. Detect ZERO $z + \text{IMPLY } x \rightarrow z$ as NOT x .
2. Keep reverse alias: $z = !x$ means $x = !z$.
3. Saves 6 pulses; final program is still FALSE/IMPLY only.

AFTER

20 steps: 4 FALSE pulses + 16 IMPLY pulses

Optimization is scheduling/alias bookkeeping, not a new hardware primitive.

